$$\rho_{a'a} = \sum_b \psi^*(a,b)\ \psi(a',b). \qquad (7.20)$$

The 2 × 2 matrix $\rho_{a'a}$ is Alice's density matrix. Notice that $\rho_{a'a}$ does not depend on any *b*-index since it has already been summed over *b*. It is purely a function of Alice variables *a* and *a'*. In fact, we only kept the *b*'s in the equation to make the example in the next section easier to follow.

We can simplify Eq. 7.19 by plugging in $\rho_{a'a}$ from Eq. 7.20. The expectation value of **L** (the 2 × 2 version) then becomes

$$\langle \mathbf{L} \rangle = \sum_{a'a} \rho_{a'a}\ L_{a,a'}. \qquad (7.21)$$

In summing over *b*, we have collapsed a 4 × 4 matrix down to a 2 × 2 matrix. This makes sense. We expect an operator that acts on the composite system to be a 4 × 4 matrix, and we expect an Alice operator to be 2 × 2.

Notice that the right side of Eq. 7.21 is a sum of diagonal matrix elements. In other words, it's the trace of the matrix $\rho\mathbf{L}$, which we can write as

$$\langle \mathbf{L} \rangle = Tr\ \rho\mathbf{L}.$$

The lesson is this: To calculate Alice's density matrix ρ, we may need to know the full wave function, including the dependence on Bob's variables. But once we know ρ, we can forget where it came from, and use it to calculate *anything* about Alice's observations. As a simple example, we can use ρ to calculate the probability $P(a)$ that Alice's system will be left in the state *a* if a measurement is performed. To determine $P(a)$, we begin with $P(a, b)$, the probability that the combined system is in state $|ab\rangle$. That's just

$$P(a,b) = \psi^*(a,b)\psi(a,b).$$

By the standard rules of probability, if we sum over *b*, we get the probability for *a*: